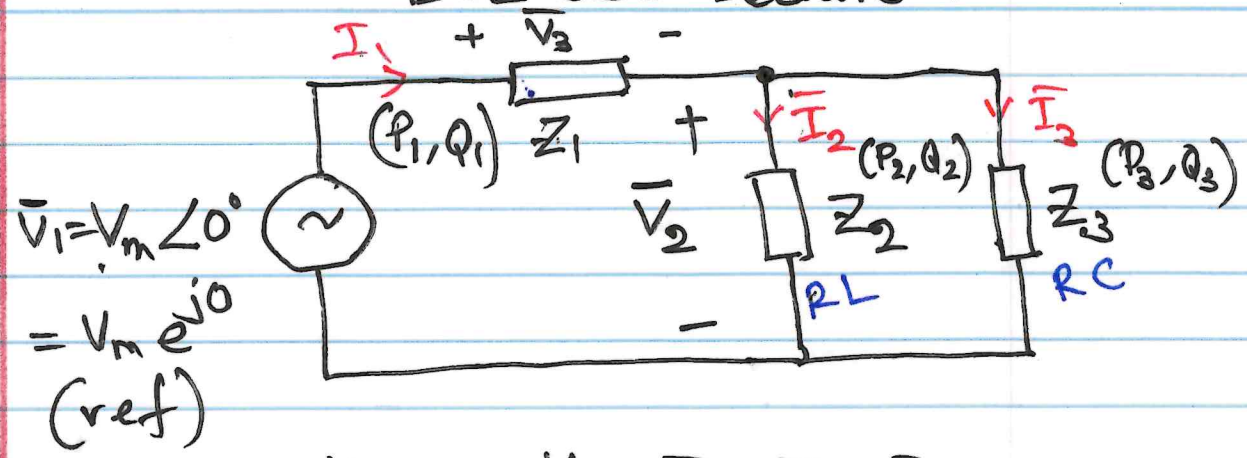
ECE 330 Lecture 4.



Known: V_m, Z_1, Z_2, Z_3

Unknown: $\bar{I}_1, \bar{I}_2, \bar{I}_3$

Power: P, Q, S

Power factor

$\bar{I}_1 = \bar{I}_2 + \bar{I}_3$ KCL

$\bar{I}_2 \bar{Z}_2 = \bar{I}_3 \bar{Z}_3$ | $\bar{V}_2 = Z_2 \bar{I}_2$

| $\bar{I}_2 = \frac{\bar{V}_2}{Z_2}$

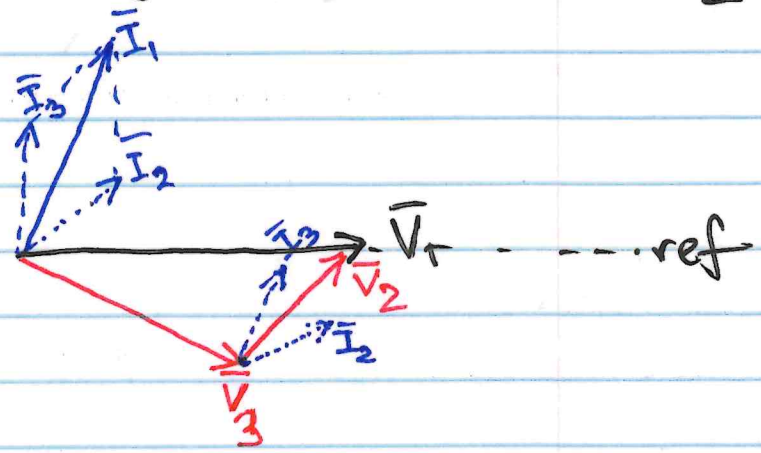
$V_m \angle 0° = \bar{V}_1 = \bar{V}_3 + \bar{V}_2$

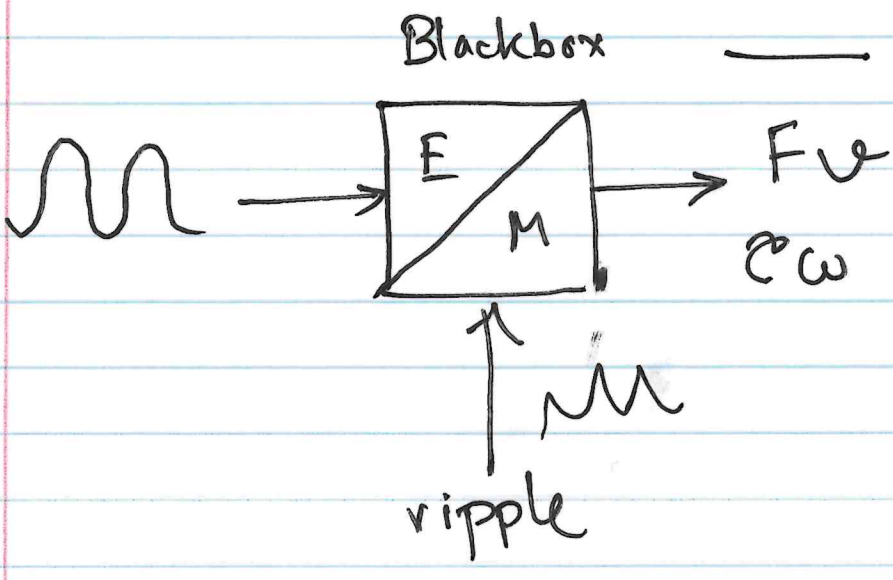
Detour:

$P = V_{rms} I_{rms} \cos\phi$

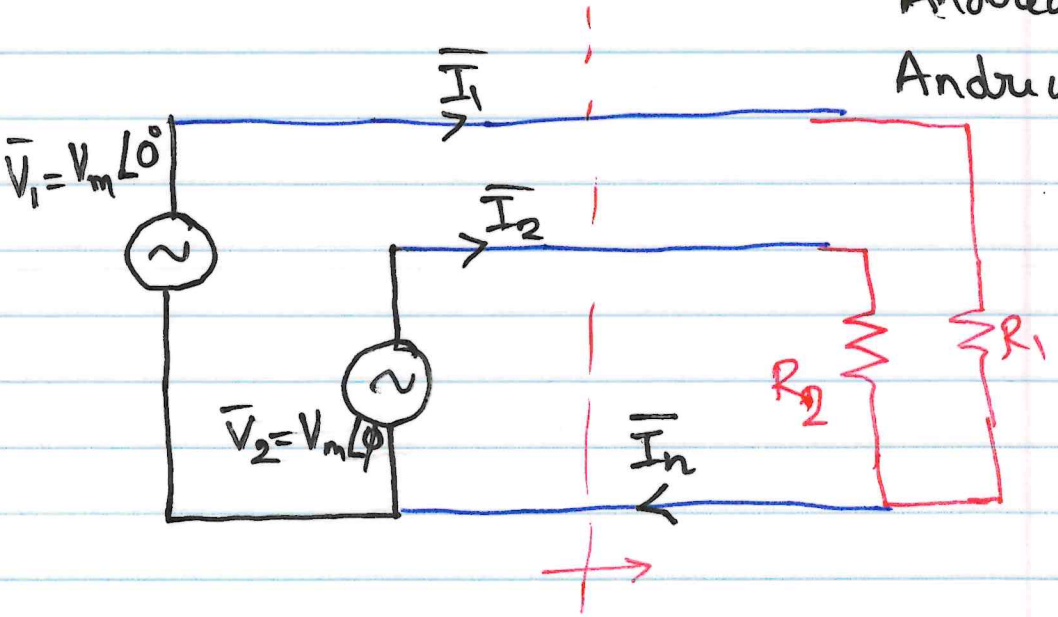
$Q = V_{rms} I_{rms} \sin\phi$

$P + jQ = \bar{V}\bar{I}^*$

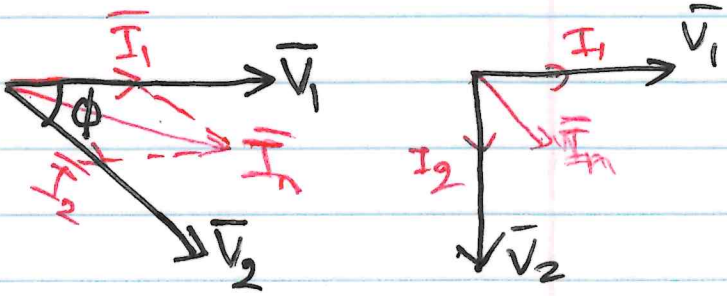




Andrew H
Andrew K.



$$\bar{I}_1 = \frac{\bar{V}_1}{R} \quad ; \quad \bar{I}_2 = \frac{\bar{V}_2}{R} .$$

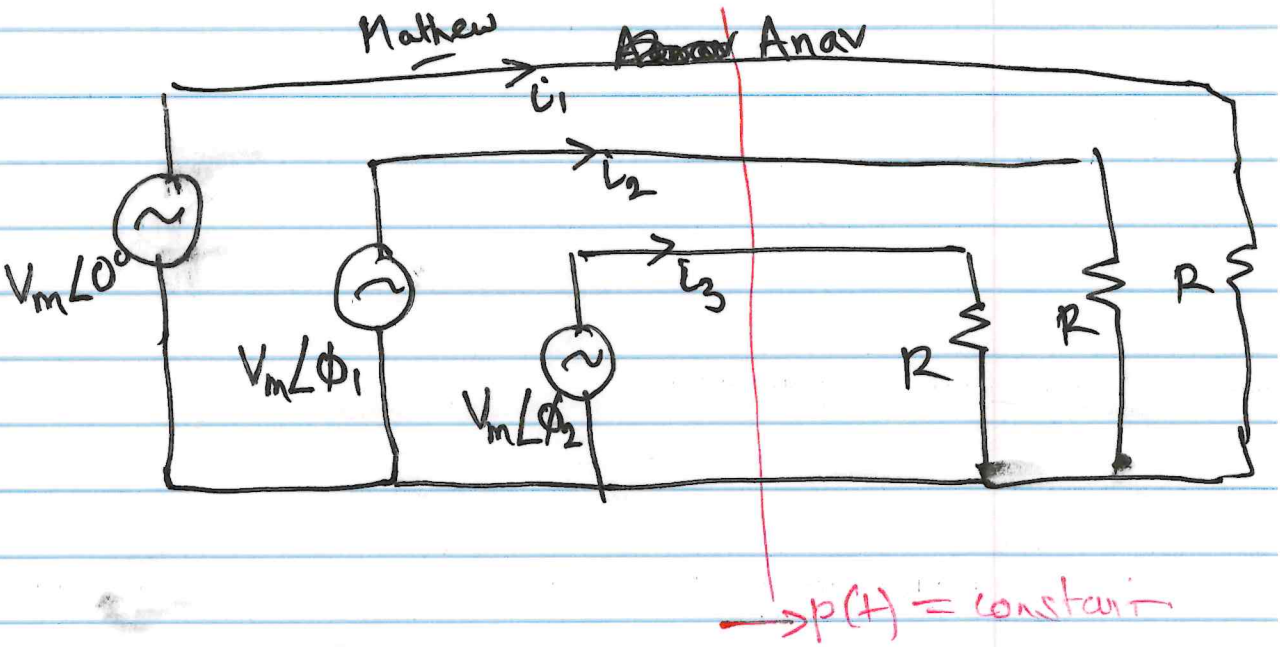


$$p_1(t) = \underbrace{2V_m \cos\omega_0 t}_{v_1} \cdot \underbrace{\frac{V_m}{R}\cos\omega_0 t}_{i_1}$$

$$p_2(t) = \underbrace{2V_m \cos(\omega_0 t - \phi)}_{v_2} \cdot \underbrace{\frac{V_m}{R}\cos(\omega_0 t - \phi)}_{i_2}$$

Nathan

$$p_1(t) + p_2(t) = K.$$



$$\bar{I}_1 + \bar{I}_2 + \bar{I}_3 = 0$$